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# CanvasTK Docs

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**Note:** You can always combine canvastk with normal tkinter. Every canvastk widget is a normal tkinter widget (a `tkinter.Canvas` or a `tkinter.Frame`), so `grid`, `pack`, `place`, `bind`, `after`, `configure`, `cget`, and so on all work.

**Note:** These are the CanvasTK docs, not the tkinter docs. Anything of tkinter will not be explained. You can find the tkinter docs [here](#).

**Note:** Most of the examples are not full code. They usually need to be preceded by `import canvastk as ctk` ; `root = ctk.Window()` and succeeded by `root.mainloop()`.

## 1 Introduction

CanvasTK is a widget toolkit for tkinter in which the widgets are drawn on a canvas instead of by the operating system. Every rounded box, border, track, thumb, tab, and popup is an anti-aliased (4x supersampled) image that is cached and shared between all widgets, so the widgets look the same everywhere, can be rounded, and can be recolored completely.

Features:

- Light and dark appearance modes and any accent color (with five ready-made ones), all switchable while the program is running.
- Every color, and also fonts, radii, border widths, and sizes, can be customized per widget.
- Large widgets are drawn from cached corner pieces, so dragging panes and resizing windows stays smooth.
- Widgets that need real text editing ([Entry](#), [Textbox](#), [Spinbox](#) and [Combobox](#)) embed a native `tkinter.Entry` or `tkinter.Text` on top of canvas chrome, so cursor, selection, clipboard, undo, and tags all behave exactly like tkinter.

## 2 Installation

### 2.1 Requirements

You need Python 3 with tkinter, and the Pillow library (`pip install pillow`). Nothing else.

### 2.2 Installing

The library is a folder (a package) called `canvastk`. To use it in one application, copy the whole `canvastk` folder next to your program. To make it available to all your programs, copy the folder into your Python libraries folder (`python3.*/site-packages`).

After that, `import canvastk` works. The version is available as `canvastk.__version__`.

## 3 Initialization

To make a window, use

```
import canvastk as ctk
root = ctk.Window()
root.mainloop()
```

`Window` is a `tkinter.Tk` whose background follows the appearance mode. If you want a subwindow of an already existing window, use `ctk.Toplevel(root)` (see [Windows](#)).

Here is a complete small program:

```
import canvastk as ctk

ctk.set_appearance_mode('dark')
ctk.set_theme(ctk.CANVAS_COLOR_THEMES['green'])

root = ctk.Window()
root.title('Hello')
ctk.Label(root, 'Your name:').grid(row = 0, column = 0, padx = 10, pady = 10)
name = ctk.Entry(root)
name.grid(row = 0, column = 1, padx = 10, pady = 10)
ctk.Button(root, 'Greet',
command = lambda: print('Hello', name.get())).grid(
row = 1, column = 0, columnspan = 2, pady = 10)
root.mainloop()
```

The file `ctk_test.py` in the project folder is a demo window showing every widget.

## 4 Concepts

### 4.1 Creating widgets

Widgets are created from the module and the *first argument is always the master*:

```
b = ctk.Button(root, 'Click Me!')
b.grid()
```

The master can be a `Window`, a `Toplevel`, a `Frame`, or any other container, including normal tkinter ones. The second argument is usually the text (or the main value) of the widget.

### 4.2 Configuring widgets

Every option listed for a widget can be given when creating it, changed later with `w.configure(option = value)` (or `w.config(...)`, or `w['option'] = value`), and read with `w.cget('option')` (or `w['option']`).

**Example:**

```
b = ctk.Button(root, 'Click Me!',
command = lambda: b.config(state = 'disabled', text = 'Done'))
b.grid()
```

Widgets redraw themselves automatically when an option changes, and the widgets that size themselves to their content (buttons, labels, check boxes, drop-downs...) resize themselves when their text, font, or image changes.

### 4.3 Size

Widgets that size to their content (`Button`, `Label`, `Checkbutton`, `Radiobutton`, `OptionMenu`) pick their own size. You can fix it by passing `width` and/or `height` in pixels. Every canvas widget, like every tkinter widget, also stretches normally when you use `sticky`, `fill`, or `expand`.

The widgets that are measured in characters and lines like in tkinter (`Entry`, `Textbox`, `Spinbox`, `Combobox`, `Listbox`) use `width` in characters and `height` in lines.

### 4.4 State

Most widgets take `state = 'normal'` or `state = 'disabled'`. Disabled widgets use their `disabled_*` colors and ignore the mouse and keyboard. `Entry` and `Combobox` also accept `'readonly'`.

## 4.5 Variables

Widgets that show or edit a value take normal tkinter variables (`tkinter.StringVar`, `IntVar`, `BooleanVar`, `DoubleVar`) and update themselves when the variable changes: the `textvariable` of a label, entry, or spinbox, the `variable` of a check box, radio button, scale, progress bar, and drop-down.

## 4.6 Colors

### 4.6.1 Named color options

Every color a widget draws is a named *part*. You can override any of them for one widget, either when creating it or later with `configure`. Passing `None` goes back to the theme color.

**Example:**

```
e = ctk.Entry(root, border_color = '#2b8a3e', focus_color = '#51cf66',
fill_color = '#f4fff6', text_color = '#1b4332')
e.grid()
e.configure(fill_color = None)    # back to the theme
```

The color options of every widget are listed in its own section. Colors can be anything tkinter understands (`'red'`, `'#ff0000'`, `'#f00'`). A color is also remembered across appearance mode changes: an override stays, everything else follows the theme.

Names that are shared between many widgets mean the same thing everywhere:

- `fill_color`, `hover_fill_color`, `disabled_fill_color`: the background of the widget in its normal, hovered, and disabled states.
- `border_color`, `hover_border_color`, `disabled_border_color`: the border.
- `focus_color`: the border/ring color while the widget has the keyboard focus.
- `text_color`, `disabled_text_color`: the text. The widgets with text also accept `fg` (the same as `text_color`).
- `arrow_color`, `disabled_arrow_color`: arrows and chevrons.
- `track_color`, `thumb_color`, `progress_color`: sliders, scroll bars, and progress bars.

### 4.6.2 The theme dictionary

Every part falls back to a theme color (see [Theme keys](#)). Instead of the named options, you can also give a **theme** dictionary of theme keys to override for that one widget. An `'accent'` in it also derives the hover color and the other keys that follow from the accent (unless you give them), just like in the global theme:

```
ctk.Button(root, 'Red accent', theme = {'accent': '#e03131'}).grid()
```

## 4.7 Fonts

Every widget that shows text takes `font`. It can be a `tkinter.font.Font` object, a font tuple, a font name string, or `None` for the default font:

```
import tkinter.font as tkfont
big = tkfont.Font(family = 'Times', size = 16, slant = 'italic', weight = 'bold')
ctk.Button(root, 'Font object', font = big).grid()
ctk.Button(root, 'Font tuple', font = ('Courier', 11, 'bold')).grid()
ctk.Label(root, 'Default font').grid()
```

`font` can be changed later with `configure`, and the widget resizes to fit. The [OptionMenu](#) and the [Combobox](#) also take `popup_font` for their drop-down lists, the [Menu](#) and the [Tooltip](#) take `font`, and the [Notebook](#) takes `font` for its tab labels.

**Note:** Fonts made from a tuple, a name, or the default are shared between widgets (making a Tk font is slow when you make hundreds of widgets). Pass your own `Font` object if you want a font you can change later; it is used as it is.

## 4.8 Images

The `image` option of a button or label takes a `PhotoImage`. Use `ctk.load_canvas_image(path, size)` to load a file (any format Pillow can read, with transparency) resized to `(width, height)`:

```
icon = ctk.load_canvas_image('save.png', (16, 16))
ctk.Button(root, 'Save', image = icon).grid()
```

Keep a reference to the image (a variable) like with any tkinter image.

## 4.9 Events

Besides the normal tkinter events, some widgets generate virtual events:

- `<<ComboboxSelected>>` — [Combobox](#), when a value is picked from the list.
- `<<ListboxSelect>>` — [Listbox](#), when the user changes the selection.
- `<<NotebookTabChanged>>` — [Notebook](#), when the selected tab changes.
- `<<NotebookTabClosed>>` — [Notebook](#), after a tab was closed.
- `<<NotebookTabMoved>>` — [Notebook](#), after the user dragged a tab to a new position.

## 4.10 Native widgets inside

`Entry`, `Textbox`, `Spinbox` and `Combobox` are frames that contain a real tkinter widget (the *inner* widget, `w.inner`). The frame forwards everything it does not have to the inner widget. So `e.get()`, `tb.insert('end', 'text')`, `tb.tag_configure(...)`, `tb.yview()`, `tb.see('end')` and so on work directly on the canvastk widget. `w.bind(...)` binds on the inner widget (except for the frame-level events `<Configure>`, `<Map>`, `<Unmap>`, `<Destroy>`, `<Enter>`, `<Leave>` and `<Visibility>`, which bind on the frame), and `w.focus_set()` focuses the inner widget.

You can also use `w.inner` directly when you need the exact tkinter widget.

## 5 Widgets

These are all the widgets:

- Button
- Label
- Checkbutton
- Radiobutton
- Entry
- Textbox
- Spinbox
- Combobox
- OptionMenu
- Menu
- MenuBar
- Scale
- Progressbar
- Scrollbar
- Separator
- Listbox
- Notebook
- Panedwindow

- GridPanedwindow
- Frame
- ScrolledFrame
- Tooltip

Now here are specifications and examples for each one:

## 5.1 Button

Use `b = ctk.Button(master, text, command, {args})` to make a button.

**Example:**

```
b = ctk.Button(root, 'Click Me!', command = lambda: print('clicked'))
b.grid()
```

The command is called when the mouse button is released over the button, or when the button has the focus and you press `Return` or `space`. While pressed, the button shrinks slightly (`press_shrink`).

Args (all of them can also be changed later with `configure`):

- `text`, `command`, `image`, `font` (default: bold, size 12), `state`.
- `default_bg`, `hover_bg`, `press_bg`: the background in the normal, hovered, and pressed states.
- `default_fg`, `hover_fg`, `press_fg`: the text color in those states.
- `disabled_bg`, `disabled_fg`: the colors when disabled.
- `border_width` (the outline width in pixels, default 1) and `border_color`.
- `radius` (the corner radius, default 15; a large radius makes a pill), `press_shrink` (default 2), `padx` (default 13) and `pady` (default 8), the space around the content.
- `focus_color`: the border color while the button has the keyboard focus, `disabled_border_color`.

Anything left as `None` follows the theme (the accent color). If you give `default_bg` but not `hover_bg`, the hover color is derived from `default_bg`: dark and medium colors get lighter on hover (a black button becomes grey), and very light colors get darker (a near-white button becomes grey), so the change is always visible (see [Hover.variant](#)). Give `hover_bg` yourself to choose it. The pressed color defaults to `default_bg` (the button just shrinks).

**Example:**

```

ctk.Button(root, 'cancel', default_bg = '#E9E2E2', hover_bg = '#F8F4F4',
press_bg = '#E9E2E2').grid(row = 0, column = 0)
ctk.Button(root, 'never', default_bg = '#FF0000', hover_bg = '#F04141',
press_bg = '#FF0000', default_fg = 'white', hover_fg = 'white',
press_fg = 'white').grid(row = 0, column = 1)

```

Method: `b.invoke()` runs the command as if the button was clicked (it briefly shows the pressed look).

## 5.2 Label

Use `l = ctk.Label(master, text, {args})` to write some text. The label is transparent (it shows the background of its master).

**Example:**

```

ctk.Label(root, 'Hello!').grid()

```

Args: `text`, `textvariable` (a `StringVar` whose value is shown), `image`, `font`, `anchor` (default 'w'), `justify`, `wraplength` (in pixels, 0 for no wrapping), `padx`, `pady` (default 2), and the colors `text_color` (or `fg`) and `disabled_text_color`.

**Example:**

```

s = tkinter.StringVar(value = 'Loading...')
ctk.Label(root, textvariable = s, font = ('TkDefaultFont', 14)).grid()
s.set('Done!')

```

## 5.3 Checkbutton

Use `c = ctk.Checkbutton(master, text, variable, onvalue, offvalue, command, {args})` to make a check box.

**Example:**

```

v = tkinter.IntVar(value = 1)
ctk.Checkbutton(root, 'Enable feature', variable = v,
command = lambda: print(v.get())).grid()

```

If you do not give a variable one is created. The default values are 1 (checked) and 0. A `BooleanVar` works too.

Methods: `c.select()`, `c.deselect()`, `c.toggle()` and `c.invoke()` (toggles and calls the command). The space bar toggles it when it has the focus.

Args: `text`, `variable`, `onvalue`, `offvalue`, `command`, `font`, `state`, and:

- `indicator_size`: the size of the box in pixels (default 18), `indicator_radius`: its corner radius, `spacing`: the space between the box and the text (default 10).
- Colors: `fill_color` (the checked box), `unchecked_color`, `hover_color` (the unchecked box when hovered), `disabled_fill_color`, `disabled_unchecked_color`, `border_color`, `hover_border_color`, `focus_color`, `disabled_border_color`, `check_color` (the check mark), `disabled_check_color`, `text_color`, `disabled_text_color`.

## 5.4 Radiobutton

Use `r = ctk.Radiobutton(master, text, variable, value, command, {args})` to make a radio button. The radio buttons that share a variable are a group.

**Example:**

```
choice = tkinter.StringVar(value = 'Red')
for name in ('Red', 'Green', 'Blue'):
    ctk.Radiobutton(root, name, variable = choice).grid(sticky = 'w')
```

The `value` defaults to the text. The args and the colors are the same as the [Checkbutton](#)'s; here `fill_color` is the color of the dot.

## 5.5 Entry

Use `e = ctk.Entry(master, {args})` to make a one-line text entry with a rounded border.

**Example:**

```
e = ctk.Entry(root, width = 30)
e.insert(0, 'Hello')
e.grid()
print(e.get())
```

Args: `placeholder`, `textvariable`, `show` (for example `'*'` for passwords), `state` (`'normal'`, `'readonly'`, `'disabled'`), `font`, `width` (in characters, default 20), `justify`, `radius` (default 8), `border_width` (default 1), `focus_border_width` (default 2), and every other option of `tkinter.Entry` (for example `validate` and `validatecommand`), which is passed to the inner entry.

Colors: `fill_color`, `disabled_fill_color`, `border_color`, `focus_color`, `disabled_border_color`, `text_color`, `disabled_text_color`, `cursor_color` (the text cursor), `selection_color`, `selection_text_color`. The `tkinter` names `bg`, `fg`, `insertbackground`, `selectbackground`, `selectforeground`, `disabledbackground`, and `disabledforeground` are accepted and mean the same.

**Placeholder.** `placeholder` is a hint text that is shown while the field is empty and disappears as soon as there is any text; it is never part of the contents (`e.get()` still returns an empty string). It is drawn in its own colors: `placeholder_color` (the text, default the muted text color) and `placeholder_fill_color` (the background of the field while the placeholder is shown; by default the normal field color). Both can be changed with `configure` and are also accepted as keys of a `theme` dictionary, and `placeholder` itself can be changed later too. It works on the [Textbox](#), [Spinbox](#), [Combobox](#) and [OptionMenu](#) as well.

**Example:**

```
search = ctk.Entry(root, placeholder = 'Search...')
```

```
placeholder_color = '#868e96')
search.grid()
```

Control-a selects all. `e.set_text(text)` replaces the text even when the entry is readonly or disabled. See [Native widgets inside](#) for the forwarded methods.

## 5.6 Textbox

Use `tb = ctk.Textbox(master, {args})` to make a multi-line text box with a rounded border.

**Example:**

```
tb = ctk.Textbox(root, wrap = 'word', width = 50, height = 12)
tb.insert('1.0', 'Some text')
tb.tag_configure('bold', font = ('TkDefaultFont', 10, 'bold'))
tb.grid()
```

Args: `width` and `height` in characters and lines (default 80 by 24, the same as tkinter's `Text`), `placeholder` (a hint shown while the text box is empty, with `placeholder_color` and `placeholder_fill_color`, exactly like the [Entry](#)'s), `scrolled` (`True` adds a built-in canvas scroll bar on the right that hides itself when not needed, default `False`), `font`, `wrap`, `state`, `width` (characters, default 40), `height` (lines, default 10), `yscrollcommand` (for a scroll bar of your own; it also works together with `scrolled`), `radius`, `border_width`, `focus_border_width`, and every other option of `tkinter.Text` (for example `undo`, `padx`, `spacing1`).

Colors: the same as the [Entry](#)'s, and for the built-in scrollbar `scrollbar_track_color`, `scrollbar_thumb_color`, and `scrollbar_thumb_hover_color`.

Control-a selects all. Everything `tkinter.Text` has (`get`, `insert`, `delete`, `index`, `see`, `tag_*`, `mark_*`, `search`, `edit_*`, `yview`...) works directly on the textbox. Use `tb.inner` for the `Text` itself.

## 5.7 Spinbox

Use `s = ctk.Spinbox(master, from_, to, increment, {args})` to make a number entry with two arrows.

**Example:**

```
s = ctk.Spinbox(root, from_ = 0, to = 10, increment = 0.5)
s.grid()
```

Args: `from_` (default 0), `to` (default 100), `increment` (default 1), `values` (a list of strings to go through instead of numbers), `wrap` (go from the last value back to the first), `command` (called with no arguments after an arrow or the Up/Down keys changed the value), `width` (default 8), and all the args of the [Entry](#) (including all its colors). The color `arrow_active_color` is used for an arrow while it is pressed.

Only text that can be the start of a valid value can be typed or pasted: digits (with a minus sign if the range goes below zero, and a decimal point with as many decimals as `increment` and the range use), or, with `values`, the beginning of one of the values. Everything else is rejected as you type. Pass your own `validatecommand` to replace this rule. Holding an arrow repeats. The Up and Down keys step too. When the entry loses the focus, the value is corrected to a number inside the range (or to one of `values`). Method: `s.step(1)` and `s.step(-1)`.

## 5.8 Combobox

Use `cb = ctk.Combobox(master, values, {args})` to make an entry with a drop-down list of values.

**Example:**

```
cb = ctk.Combobox(root, values = ['Python', 'LaTeX', 'Plain text'],
state = 'readonly')
cb.set('Python')
cb.bind('<<ComboboxSelected>>', lambda e: print(cb.get()))
cb.grid()
```

Args: `values`, `command` (called with the picked value), `postcommand` (called, with no arguments, right before the list opens, so it can update `values`), `state` ('readonly' makes the whole field open the list instead of being editable), `popup_font`, `popup_max_height`, and all the args of the [Entry](#).

The drop-down list does not grow to the bottom of the screen: it is at most `popup_max_height` pixels tall (default 260), and a longer list scrolls. It has a scroll bar that you can drag (click the track to page), and you can also use the mouse wheel and the Up, Down, Home, and End keys. The list opens scrolled so that the current value is in view. `cb.configure(popup_max_height = 150)` changes the limit later.

Methods: `cb.set(value)`, `cb.get()`, `cb.current()` (the index of the value, or -1) and `cb.current(index)` (sets it). The Down key opens the list.

Colors: all the [Entry](#)'s, and for the drop-down list: `popup_fill_color`, `popup_border_color`, `popup_hover_color` (the hovered row), `popup_hover_text_color`, `popup_text_color`, `popup_disabled_text_color`, `popup_muted_text_color`, `popup_current_color` (the row of the current value), `popup_separator_color`, `popup_thumb_color` (the scroll indicator).

## 5.9 OptionMenu

Use `o = ctk.OptionMenu(master, variable, default, *values, {args})` to make a drop-down button for choosing one of some values.

**Example:**

```
v = tkinter.StringVar()
o = ctk.OptionMenu(root, v, 'Light', 'Light', 'Dark',
```

```
command = lambda value: print('chose', value))
o.grid()
```

The `default` is selected at the start and added to the values if it is missing. Pass `None` as the variable to let the widget make its own one (`o.variable`).

Args: `command` (called with the picked value), `placeholder` (text shown while nothing is chosen, that is while the variable is empty, in `placeholder_color`, with `placeholder_fill_color` as the background; the same options as the `Entry`'s), `radius`, `border_width`, `focus_border_width`, `popup_font`, `popup_max_height`, `font`, `state`.

The list is at most `popup_max_height` pixels tall (default 260); a longer list scrolls, exactly like the `Combobox`'s (draggable scroll bar, mouse wheel, Up/Down/Home/End, opening at the current value). The `values` list can be changed later: `o.configure(values = [...])`.

Colors: `fill_color`, `hover_fill_color`, `disabled_fill_color`, `border_color`, `hover_border_color`, `focus_color`, `disabled_border_color`, `text_color`, `disabled_text_color`, `arrow_color`, `disabled_arrow_color`, and all the drop-down list colors (`popup_*`) of the `Combobox`.

## 5.10 Menu

Use `m = ctk.Menu(master, {args})` to make a popup menu. It is not a widget, it is an object that opens a popup (with submenus) where you tell it to.

**Example:**

```
m = ctk.Menu(root)
m.add_command(label = 'New', accelerator = 'Ctrl+N',
command = lambda: print('new'))
m.add_command(label = 'Disabled', state = 'disabled')
m.add_separator()
recent = ctk.Menu(root)
recent.add_command(label = 'notes.txt', command = lambda: print('notes'))
m.add_cascade(label = 'Recent', menu = recent)
wrap = tkinter.BooleanVar(value = True)
m.add_checkbutton(label = 'Word wrap', variable = wrap)
root.bind('<Button-3>', lambda e: m.post(e.x_root, e.y_root))
```

Adding entries (the `insert_*` versions take the position as their first argument):

- `add_command(label, command, accelerator, state)`
- `add_separator()`
- `add_checkbutton(label, variable, onvalue, offvalue, command, accelerator, state)`
- `add_radiobutton(label, variable, value, command, accelerator, state)`
- `add_cascade(label, menu, state)`

Other methods: `m.post(x, y)` (or `tk.popup`) shows the menu at screen coordinates, `m.unpost()`, `m.delete(first, last)`, `m.index(label_or_position)`, `m.entryconfigure(index, **options)`, `m.entrycget(index, option)` and `m.invoke(index)`. Positions can be numbers, 'end', or a label.

The popup closes when you click outside it, press Escape, move the window, or pick an entry. You can use the mouse, or Up, Down, Home, End, Return, Right (open a submenu), Left and Escape.

Colors (arguments of `Menu`, or `m.configure(...)`): `fill_color`, `border_color`, `hover_color`, `hover_text_color`, `text_color`, `disabled_text_color`, `muted_text_color` (the accelerators), `current_color`, `separator_color`, `thumb_color`. The font is an argument too.

**Note:** The popup is a borderless window, so its corners are square.

## 5.11 MenuBar

Use `mb = ctk.MenuBar(master, menu)` to make a bar like a window's menu bar. It shows the cascade entries of the [Menu](#) you give it.

**Example:**

```
bar = ctk.Menu(root)
file_menu = ctk.Menu(root)
file_menu.add_command(label = 'Quit', command = root.destroy)
bar.add_cascade(label = 'File', menu = file_menu)
ctk.MenuBar(root, bar).pack(fill = 'x')
```

Clicking a name opens its menu below it; while a menu is open, hovering another name switches to it. The bar updates itself when you add or remove cascades from the menu. Use `mb.set_menu(menu)` to show another menu.

Colors: `hover_color`, `open_color`, `text_color`, `disabled_text_color`.

## 5.12 Scale

Use `s = ctk.Scale(master, from_, to, {args})` to make a slider.

**Example:**

```
s = ctk.Scale(root, from_ = 0, to = 100, length = 300,
command = lambda v: print(v))
s.set(35)
s.grid()
```

Args: `from_`, `to`, `orient` ('horizontal' or 'vertical'), `variable` (a `DoubleVar` is made if not given), `command` (called with the new value as a number), `resolution` (the step; 0 means any value), `length` (in pixels, default 160), `value` (the starting value), `track_thickness` (default 6), `thumb_size` (default 18).

Methods: `s.get()` and `s.set(value)`. You can click or drag, use the mouse wheel, the arrow keys, Home, and End.

Colors: `track_color`, `progress_color` (the filled part),

disabled\_progress\_color, thumb\_color, thumb\_hover\_color,  
disabled\_thumb\_color, thumb\_border\_color, thumb\_hover\_border\_color,  
disabled\_thumb\_border\_color, focus\_color.

### 5.13 Progressbar

Use `p = ctk.Progressbar(master, {args})` to make a progress bar.

**Example:**

```
p = ctk.Progressbar(root, length = 300)
p.configure(value = 40)
p.grid()
busy = ctk.Progressbar(root, length = 300, mode = 'indeterminate')
busy.grid()
busy.start()
```

Args: orient, length (default 200), mode ('determinate' or 'indeterminate'), maximum (default 100), value, variable, track\_thickness (default 10). Colors: track\_color and progress\_color. Methods: `p.start(interval = 15)` switches to the indeterminate mode and animates a bouncing bar, `p.stop()`, and `p.step(amount = 1)` adds to the value (and wraps around the maximum).

### 5.14 Scrollbar

Use `sb = ctk.Scrollbar(master, command, {args})` to make a scroll bar. It connects to widgets exactly like the tkinter one.

**Example:**

```
lb = ctk.Listbox(root, ['Item %d' % n for n in range(100)])
sb = ctk.Scrollbar(root, command = lb.yview)
lb.configure(yscrollcommand = sb.set)
lb.grid(row = 0, column = 0)
sb.grid(row = 0, column = 1, sticky = 'ns')
```

Args: orient ('vertical' or 'horizontal'), command, autohide (draw nothing while everything is visible, default `False`), thickness (default 14). You can drag the thumb, click the track to page, and use the mouse wheel on it. Methods: `sb.set(first, last)` and `sb.get()`. Colors: track\_color, thumb\_color, thumb\_hover\_color, disabled\_thumb\_color.

### 5.15 Separator

Use `ctk.Separator(master, orient, thickness)` to draw a line (fill = 'x' for a horizontal one, fill = 'y' for a vertical one).

**Example:**

```
ctk.Separator(root).pack(fill = 'x', padx = 10, pady = 8)
```

Args: `orient` (default 'horizontal'), `thickness` (default 1), and the color color.

## 5.16 Listbox

Use `lb = ctk.Listbox(master, contents, {args})` to make a list that you can select from. `contents` is a list (or tuple) of the items.

**Example:**

```
lb = ctk.Listbox(root, ['One', 'Two', 'Three'], height = 5,
selectmode = 'extended')
lb.bind('<<ListboxSelect>>', lambda e: print(lb.curselection()))
lb.grid()
```

Args: `width` (characters, default 20), `height` (rows, default 10), `selectmode` ('browse', 'single', 'multiple', or 'extended'; extended supports Ctrl-click, Shift-click, and dragging), `yscrollcommand`, `scrolled` (`True` draws a scroll bar inside the box, on the right, that only shows while the list is longer than the box; you can drag the thumb and click the track to page; default `False`), `font`, `radius`, `border_width`, `focus_border_width`, `row_padding` (extra pixels of height per row, default 8), `row_radius` (the corner radius of the selection highlight, default 5).

With `scrolled` you do not need a separate [Scrollbar](#); `yscrollcommand` still works if you also want to connect a scroll bar of your own.

The methods are the ones of the tkinter `Listbox`: `insert`, `delete`, `get`, `size`, `index`, `curselection`, `selection_get` (the text of the selected rows, one per line), `selection_set`, `selection_clear`, `selection_includes`, `selection_anchor`, `activate`, `nearest`, `see`, `yview`, `yview_moveto`, `yview_scroll`. The keys Up, Down, Page Up, Page Down, Home, End, and the mouse wheel work. Double clicks can be bound normally with `<Double-Button-1>`.

Colors: `fill_color`, `disabled_fill_color`, `border_color`, `focus_color`, `disabled_border_color`, `text_color`, `disabled_text_color`, `hover_color`, `selected_color` (the selected rows while the list has the focus), `selected_inactive_color`, `selected_text_color`, and for the built-in scroll bar `scrollbar_track_color`, `scrollbar_thumb_color`, and `scrollbar_thumb_hover_color`.

## 5.17 Notebook

Use `nb = ctk.Notebook(master, {args})` to make tabs. Make each page with the notebook as its master, then add it with `add`.

**Example:**

```
nb = ctk.Notebook(root)
nb.pack(fill = 'both', expand = True, padx = 10, pady = 10)
```

```

page1 = ctk.Frame(nb)
ctk.Label(page1, 'Page one').pack(padx = 20, pady = 20)
nb.add(page1, text = 'First')
page2 = ctk.Frame(nb)
ctk.Button(page2, 'Page two').pack(padx = 20, pady = 20)
nb.add(page2, text = 'Second')
nb.bind('<<NotebookTabChanged>>', lambda e: print(nb.index('current'))))

```

The notebook sizes itself to its biggest page, and the pages of a `Frame` take the page background color automatically.

**Styles.** `style = 'segmented'` (the default) draws a rounded segmented control (like macOS or customtkinter's tab view) above a rounded page, with the selected tab filled with the accent color. `style = 'classic'` draws folder-like tabs joined to the page:

```
nb = ctk.Notebook(root, style = 'classic')
```

**Closable tabs.** `closable = True` puts a close button on every tab. A single tab can override it with its own `closable`:

```

nb = ctk.Notebook(root, closable = True,
on_close = lambda page: print('closing', page))
nb.add(page1, text = 'First', closable = False)

```

Clicking the `×` (or middle-clicking the tab) calls `on_close(page)`; if it returns `False` the tab stays. Otherwise the tab is removed, and its page is destroyed unless you pass `destroy_on_close = False`. Then `<<NotebookTabClosed>>` is generated.

**Draggable tabs.** `draggable = True` lets the user reorder the tabs by dragging them along the strip (press a tab, move the mouse a few pixels, and drag; the cursor changes while dragging). The tabs swap places live as the dragged tab passes the middle of its neighbor, the selected tab stays selected, and when the strip overflows, dragging near its ends scrolls it. `<<NotebookTabMoved>>` is generated when the tab is dropped. It works in both styles, together with closable tabs.

**Many tabs.** When the tabs do not fit, an arrow appears at each end of the strip (`<` on the left, `>` on the right) (they highlight when hovered and dim at the ends); each click brings the next hidden tab into view. The mouse wheel over the strip scrolls it too.

Args: `font`, `radius` (default 8; the corner radius of the page, and of the tabs in the classic style), `segment_radius` (the corner radius of the tabs in the segmented style; the default is 10, round corners; a smaller number such as 6 gives softer corners, 0 is square, and `None` makes fully rounded pills; the pill behind the tabs follows it), `padding` (space around the page, default 8), `tab_padx` (default 16), `closable`, `on_close`, `destroy_on_close`, `draggable` (default `False`), `style`. All of them except `style` can be changed later with `configure`.

Methods (tabs can be given as a page widget, a number, `'current'`, or `'end'`):

- `nb.add(page, text, state, image, closable)` and `nb.insert(when, page, ...)`. `state` is 'normal', 'disabled', or 'hidden'.
- `nb.forget(tab)` removes a tab without destroying the page, `nb.hide(tab)` hides it.
- `nb.move(tab, position)` moves a tab to a new position (a number or 'end') from your code; the selected tab stays selected.
- `nb.select()` returns the selected page, `nb.select(tab)` selects one.
- `nb.tab(tab, option)` reads and `nb.tab(tab, option = value)` changes the text, state, image, or closable of a tab.
- `nb.tabs()` (the pages), `nb.index(tab)`, `nb.close_tab(tab)`, `nb.is_closable(tab)`.
- `nb.enable_traversal()` makes Control-Tab and Control-Shift-Tab switch tabs. The Left and Right keys switch tabs while the notebook has the focus.
- `nb.update_size()` recalculates the requested size after you changed the contents of a page.

Colors: `tab_color`, `tab_hover_color`, `tab_text_color`, `disabled_tab_text_color`, `selected_tab_color`, `selected_tab_text_color`, `border_color`, `page_color` (the page and the background of its Frames), `track_color` (the pill behind the segmented tabs), `close_color`, `close_hover_color` (the × turns red when hovered), `arrow_color`, `arrow_hover_color`, `arrow_hover_text_color`.

## 5.18 Panedwindow

Use `pw = ctk.Panedwindow(master, orient)` to make resizable panes separated by draggable sashes (with a grip drawn on each). Make the panes with the panedwindow as their master.

**Example:**

```
pw = ctk.Panedwindow(root, orient = 'horizontal')
pw.pack(fill = 'both', expand = True)
pw.add(ctk.Listbox(pw, ['a', 'b', 'c']), weight = 0)
right = ctk.Panedwindow(pw, orient = 'vertical')
pw.add(right, weight = 1)
right.add(ctk.Textbox(right), weight = 1)
right.add(ctk.Button(right, 'Bottom pane'), weight = 1)
```

The `weight` says how the panes share extra space when the window is resized (a pane with weight 0 keeps its size; if all weights are 0 the last pane gets the

extra space). A pane starts with the size it asks for and cannot become smaller than 24 pixels.

Methods: `pw.add(pane, weight)`, `pw.insert(where, pane, weight)`,  
`pw.forget(pane)` (or `remove`), `pw.panes()`,  
`pw.paneconfigure(pane, weight = 2)` (or `pane`), `pw.sashpos(index)`  
reads and `pw.sashpos(index, position)` sets the position of a sash.

Colors: `sash_color`, `sash_active_color` (hovered or dragged), `grip_color`,  
`grip_active_color`.

## 5.19 GridPanedwindow

Use `gp = ctk.GridPanedwindow(master)` to make resizable panes that can be split in *both* directions inside one widget. A [Panedwindow](#) only goes one way; to split one of its panes the other way you have to nest a second panedwindow in it, and every nested panedwindow is a separate widget. In a `GridPanedwindow` you take any pane, however deep, and split it horizontally or vertically, and then split the new panes again.

**Example:**

```
gp = ctk.GridPanedwindow(root)
gp.pack(fill = 'both', expand = True)

files = ctk.Listbox(gp, ['main.py', 'util.py'], scrolled = True)
gp.add(files)                                # the first pane

editor = ctk.Textbox(gp)
gp.split(files, editor, 'horizontal')        # files / editor

output = ctk.Textbox(gp)
gp.split(editor, output, 'vertical')         # editor above output

terminal = ctk.Textbox(gp)
gp.split(output, terminal, 'horizontal')     # output / terminal
```

This makes a list on the left, with a text box over two side-by-side text boxes on the right. The last line splits the *bottom* pane sideways, which is exactly what one panedwindow cannot do.

**How it works.** The panes form a tree of rows and columns. Every row or column (a *split*) has its own sashes between its panes, and dragging a sash moves only the border between its two neighbors. The panes are the widgets you give it, created with the `GridPanedwindow` as their master (also the ones deep in the layout); the widget places them itself, so do not `pack` or `grid` them. Each pane keeps its proportion of the space when the window is resized, and no pane can be made smaller than 24 pixels by dragging.

**Junctions.** Where a sash ends on a sash of the other direction (for example, a horizontal sash in the right column meeting the vertical sash between the columns), the two meet at a *junction*. Press and drag near that point, within

about 9 pixels, and every sash that meets there moves together: the vertical one sideways and the horizontal one up or down at the same time. The pointer becomes a four-way move cursor and all the sashes of the junction light up while you are near it. If the panes on both sides have sashes that line up, they all move together. Dragging a sash anywhere else along its length moves only that sash, as before.

Methods:

- `gp.add(pane)` adds the first pane. Later, `gp.add(pane, orient)` appends a pane to the whole layout: to the right of everything ('horizontal', the default) or below everything ('vertical'), sharing the space equally with the other top-level panes.
- `gp.split(target, pane, orient, before = False, ratio = 0.5)` splits the pane `target` (a widget that is already in the layout) in two: `pane` goes to its right ('horizontal') or below it ('vertical'), or to its left or above it with `before = True`. `ratio` is the part of the target's space that the new pane gets. Only the target's own space is divided.
- `gp.remove(pane)` (or `forget`) takes a pane out of the layout and gives its space to its neighbors; the widget is not destroyed. A row or column left with one pane disappears.
- `gp.panes()` returns all the panes in layout order.

**Example:**

```
terminal_pane = gp.panes()[-1]
gp.remove(terminal_pane)
terminal_pane.destroy()
```

Colors: `sash_color`, `sash_active_color` (hovered or dragged), `grip_color`, `grip_active_color`, like the [Panedwindow](#).

## 5.20 Frame

Use `f = ctk.Frame(master, {args})` to make a frame. It is a normal `tkinter.Frame` that follows the appearance mode: it has the same background as its master (a notebook page, a window...).

**Example:**

```
f = ctk.Frame(root)
f.pack(padx = 10, pady = 10)
ctk.Label(f, 'Inside a frame').pack()
```

Give `bg` for a fixed color that does not follow the theme. All other args are the same as `tkinter`'s.

## 5.21 ScrolledFrame

Use `sf = ctk.ScrolledFrame(master, {args})` to make a frame that scrolls. Put your widgets inside the `ScrolledFrame` itself; `pack`, `grid`, and `place` act on the whole scrolling container.

**Example:**

```
sf = ctk.ScrolledFrame(root, orient = 'vertical')
sf.pack(fill = 'both', expand = True)
for n in range(50):
    ctk.Checkbutton(sf, 'Option %d' % n).pack(anchor = 'w')
```

Args: `orient` ('vertical' (default), 'horizontal' or 'both'), `autohide` (the scroll bars hide themselves when not needed, default `True`), `bg`, and the scroll bar colors `scrollbar_track_color`, `scrollbar_thumb_color`, `scrollbar_thumb_hover_color`.

The mouse wheel scrolls vertically (horizontally for 'horizontal'), and Shift with the wheel always scrolls horizontally. Text boxes and list boxes inside keep their own wheel scrolling.

**Example:**

```
wide = ctk.ScrolledFrame(root, orient = 'horizontal')
wide.pack(fill = 'x')
for n in range(30):
    ctk.Button(wide, 'Button %d' % n).pack(side = 'left', padx = 6, pady = 10)
```

## 5.22 Tooltip

Use `ctk.Tooltip(widget, text, {args})` to show a popup text when the mouse rests on a widget.

**Example:**

```
b = ctk.Button(root, 'hover me')
b.grid()
ctk.Tooltip(b, 'A canvas-drawn tooltip')
```

Args: `delay` (in milliseconds before it shows, default 500), `font`, `fill_color`, `border_color`, `text_color`. The text can have several lines. It hides when the mouse leaves or is pressed. It does not replace the widget's own bindings.

# 6 Windows

## 6.1 Window

`root = ctk.Window()` makes the application window. It is a `tkinter.Tk`, so everything such as `title`, `geometry`, `minsize`, `protocol`, and `mainloop` works. Its background follows the appearance mode, unless you give `bg` a color such as `'#ffffff'` (then it stays that color).

Args: `bg` and `theme`, a dictionary of theme keys that applies to the window and to every widget inside it:

```
root = ctk.Window(theme = {'window': '#14213d', 'surface': '#1d2d50',
                           'text': '#ffffff', 'accent': '#fca311'})
```

Methods (see [Changing the theme dynamically](#)):

- `root.set_theme(theme)` changes theme keys for the window and all the widgets in it, immediately. A value of `None` removes a change.
- `root.get_theme()` returns the window's current theme as a dictionary.
- `root.reset_theme()` removes every change made with `theme` or `set_theme`.

## 6.2 Toplevel

`top = ctk.Toplevel(root)` makes a subwindow, with the same arguments and methods. By default it is a real transient subwindow of its master (`transient = True`): the window manager keeps it grouped with the main window (a single taskbar entry, minimized and raised together with it, and shown above it). Pass `transient = False` for a completely independent window. Its theme is independent of the main window's:

```
top = ctk.Toplevel(root, theme = ctk.CANVAS_DARK_PALETTE)    # a dark window
```

## 7 File dialogs

`canvastk.filedialog` is a drop-in replacement for `tkinter.filedialog`: the same functions, classes, options, and return values, but drawn with `canvastk` widgets, so it matches your application (and follows the theme of the window it belongs to) instead of the system dialog.

**Example:**

```
from canvastk import filedialog

path = filedialog.askopenfilename(
    title = 'Open a text file',
    filetypes = [('Text files', '*.txt'), ('Python files', '*.py *.pyw'),
                 ('All files', '*')])

if path:
    print('opening', path)

paths = filedialog.askopenfilenames()                # a tuple of paths
target = filedialog.asksaveasfilename(defaultextension = '.txt')
folder = filedialog.askdirectory(mustexist = True)
with filedialog.asksaveasfile(mode = 'w') as handle: # an open file, or None
    handle.write('hello')
```

## 7.1 Functions

- `askopenfilename(**options)`: choose an existing file; returns its path, or `''` if cancelled.
- `askopenfilenames(**options)`: choose several files; returns a tuple of paths (empty if cancelled).
- `asksaveasfilename(**options)`: choose where to save; returns the path, or `''` if cancelled.
- `askdirectory(**options)`: choose a folder; returns its path, or `''` if cancelled.
- `askopenfile(mode = 'r', **options)`, `askopenfiles(mode = 'r', **options)`, and `asksaveasfile(mode = 'w', **options)`: the same, but they return the file(s) opened in `mode` (a list for `askopenfiles`), or `None` if cancelled.
- The classes `Open`, `SaveAs`, and `Directory`: `Open(master, **options).show(**more_options)` does what the matching function does. Options given to `show` override the ones given to the constructor.

## 7.2 Options

- `parent`: the window the dialog belongs to (the default window if omitted). The dialog is modal, centered on it, and grouped with it by the window manager.
- `title`: the dialog's title (`'Open'`, `'Save As'`, or `'Select Folder'` by default).
- `initialdir`: the folder to start in. By default the folder the previous file dialog ended in, or the current folder.
- `initialfile`: the file name to fill in (it may include a folder, which then becomes the starting folder; in an open dialog the file is also selected).
- `filetypes`: a list of (`label`, `patterns`). `patterns` is a string such as `'*.png *.jpg'` or a sequence of patterns; a pattern may be a glob (`'*.txt'`), an extension (`'*.txt'` or `'txt'`), or `'*'` for everything. The first entry is selected at the start. Without `filetypes` there is a single `'All files'` entry.
- `typevariable`: a `StringVar` that gives the file type to start with (by its label) and receives the label of the type that was selected when the dialog is accepted.
- `defaultextension`: added to the name typed in a save dialog if it has no extension.

- **multiple**: allow choosing several files (`askopenfilenames` sets it).
- **mustexist**: for `askdirectory`, whether the folder must already exist (a typed name that does not exist is refused).
- **confirmoverwrite**: in a save dialog, ask before replacing an existing file (default `True`).

An unknown option raises a `TypeError`, like an unexpected keyword argument. The `message` option of `askdirectory` is accepted and ignored.

### 7.3 Using the dialog

- **Places** on the left jump to your home folder, its usual folders (Desktop, Documents, Downloads...), mounted drives, and the file system root.
- The **back**, **forward**, and **up** buttons move through the folders you visited. The **path bar** shows the folder as a row of clickable pieces: click any folder in it to go there (an easy way out of where you are; a long path collapses its first folders into a “...” piece). Click the empty space after the last piece, or press Control-L, to type a path instead and press Return (a path to a file opens its folder and selects the file); Escape goes back to the bar.
- The file list shows an icon, **Name**, **Size**, and **Modified**. Click a column title to sort by it, and click it again to reverse the order; folders always come first. Select with the mouse or the arrow keys (Page Up, Page Down, Home, End); Ctrl-click and Shift-click select several files when **multiple** is set. Typing letters jumps to the first name that starts with them. Double click opens a folder or accepts a file; Return accepts.
- The **Name** field holds the selected name (or the names, quoted if they contain spaces); in an open dialog you can also type a name or a path, and in a save dialog the name of the new file. While you type there, the list is narrowed to the names that contain the text, and the rest of a name that begins with the text is suggested as selected text after it. Tab or Return accepts the suggestion (Return with no suggestion showing accepts the dialog as usual), Escape hides it until you type again, and typing on replaces it. When several names fit what you typed, a dropdown lists them under the field: Up and Down move through it, a click or Return picks the highlighted name, and Escape hides it. Picking a file only puts its name in the field (so you can still change it before accepting); picking a folder enters it. If you type a path (such as `Dir/file`, `/file`, or `~/file`), the list is not narrowed and the last piece is suggested from the files and folders of that folder. The path field suggests folder names the same way (Tab or Return accepts, Escape hides; Return with no suggestion opens the path). The **Type** box filters the files by the **filetypes** (folders are always shown). **Hidden** shows hidden files.

- In save and folder dialogs (not in open dialogs), **New folder** (or the right click menu) makes a folder in the current one.
- Keyboard shortcuts: Escape cancels, Alt-Left, Alt-Right, and Alt-Up go back, forward, and up, F5 refreshes, Control-h shows or hides hidden files, and Control-l edits the path.
- The open dialog refuses names that are not files, the save dialog asks before replacing a file, and a folder that cannot be read shows an error instead of closing the dialog.

## 7.4 Helper functions

The logic that does not need a window is available as plain functions (useful for your own file pickers and for testing):  
`parse_filetypes(filetypes)`, `name_matches(name, patterns)`,  
`list_directory(directory, show_hidden, patterns, directories_only)`  
 (returns the entries and an error), `sort_entries(entries, column, descending)`,  
`default_places()`, `resolve_path(text, directory)`,  
`with_default_extension(name, extension)`, `human_size(size)`,  
`format_modified(timestamp)`, `filter_entries(entries, text)`,  
`complete_name(names, text)`, and `complete_path(text, directory, show_hidden, directories_only)`.

## 8 Message boxes

`canvastk.messagebox` is a drop-in replacement for `tkinter.messagebox`: the same functions, constants, options, and return values, drawn with `canvastk` widgets. A message box is modal, follows the theme of the window it belongs to, and is centered over it. The file dialogs use it for their errors and questions.

**Example:**

```
from canvastk import messagebox

messagebox.showinfo('Saved', 'The file was saved.')
messagebox.showerror('Error', 'Could not open the file.', detail = 'Permission denied.')
if messagebox.askyesno('Quit', 'Quit without saving?'):
    root.destroy()
answer = messagebox.askyesnocancel('Save', 'Save the changes?') # True, False, or None
```

|                                                                                |                                                                |
|--------------------------------------------------------------------------------|----------------------------------------------------------------|
| <code>showinfo</code> , <code>showwarning</code> , <code>showerror</code>      | return 'ok'                                                    |
| <code>askquestion</code>                                                       | returns 'yes' or 'no'                                          |
| <code>askokcancel</code> , <code>askyesno</code> , <code>askretrycancel</code> | return <b>True</b> (OK, Yes, Retry) or <b>False</b>            |
| <code>askyesnocancel</code>                                                    | returns <b>True</b> , <b>False</b> , or <b>None</b> for Cancel |

All of them take (title, message, **\*\*options**). The options are **parent** (the window it is shown over), **default** (the button that Return presses, such as 'no'), **detail** (a second, smaller text under the message), **buttons** (your own button labels), **icon**

(`messagebox.ERROR`, `INFO`, `QUESTION`, or `WARNING`), and `type` (`OK`, `OKCANCEL`, `YESNO`, `YESNOCANCEL`, `RETRYCANCEL`, or `ABORTRETRYIGNORE`). A list of your own buttons can be given with `buttons`, as in `messagebox.askquestion('Delete', 'Delete the file?', buttons = ['Delete', 'Keep'], default = ...` the buttons are shown in that order, and the function returns the label of the pressed button as a string (the `ask` functions that normally return a boolean return the label too, since your labels have no yes or no meaning). `default` is then one of your labels, and closing the window (or pressing Escape) returns `None`. An unknown option raises a `TypeError`. Closing the window or pressing Escape answers Cancel, or No when there is no Cancel button.

## 9 Color picker

`canvastk.colorchooser` is a drop-in replacement for `tkinter.colorchooser`, drawn with `canvastk` widgets: the same `askcolor` function and `Chooser` class, the same options, and the same return value. The dialog is modal, follows the theme of the window it belongs to, and is centered over it.

**Example:**

```
from canvastk import colorchooser

rgb, hex_color = colorchooser.askcolor('#3b82f6', title = 'Pick a color')
if hex_color:
    # (None, None) when the dialog was cancelled
    print(rgb, hex_color)      # (59, 130, 246) #3b82f6
```

`askcolor(color = None, **options)` returns `((r, g, b), '\#rrggbb')` with the three numbers from 0 to 255 and a lowercase hex code, or `(None, None)` if the dialog was cancelled or closed. `colorchooser.Chooser(master, **options).show()` does the same as a class. The color to start from is the first argument or the `initialcolor` (or `color`) option, and can be a hex code (`'#3b82f6'` or `'#38f'`), a string of three numbers (`'59, 130, 246'`), a tuple, or a Tk color name; the default is white. The other options are `parent` (the window the dialog is shown over; the default is the main window) and `title`. An unknown option raises a `TypeError`, and a color that cannot be understood raises a `ValueError`.

### 9.1 Using the dialog

- The **square** chooses the saturation (left to right) and the brightness (top to bottom) of the current hue; press and drag in it. The **rainbow bar** under it chooses the hue.
- The **Hex** field and the **R**, **G**, and **B** fields show the color as numbers, and you can type in them: the rest of the dialog follows as soon as what you typed is a valid color. The three number fields also step with the arrow buttons and the Up and Down keys.

- The **Colors** palette has twenty-four ready-made colors, and **Recent** lists the colors you chose before, newest first (twelve at most, kept until the program ends). Click a swatch to choose it; the swatch of the current color has an accent-colored ring.
- Return chooses the color (OK) and Escape cancels, like the two buttons.

## 9.2 Helper functions

The conversions that do not need a window are available as plain functions: `parse_color(text)` (the (r, g, b) of a hex code or of 'r, g, b', or `None`), `rgb_to_hex(rgb)`, `rgb_to_hsv(rgb)`, `hsv_to_rgb(hue, saturation, value)` (hue, saturation, and value are numbers from 0 to 1), `square_image(hue, width, height)`, and `hue_strip_image(width, height)` (the gradients as Pillow images).

# 10 Appearance and themes

## 10.1 Appearance modes

`ctk.set_appearance_mode(mode)` sets the mode to 'light' or 'dark'. Every existing widget, window, and popup recolors immediately, and the mode is used for everything made afterwards. `ctk.get_appearance_mode()` returns 'light' or 'dark'.

**Example:**

```
ctk.OptionMenu(root, None, 'Light', 'Light', 'Dark',
command = ctk.set_appearance_mode).grid()
```

## 10.2 Accent color

The accent color is used by buttons, selected tabs, check boxes, slider fills, selections, and so on. It is a theme key like any other, so you set it with `ctk.set_theme` (see [Changing the theme dynamically](#)):

```
ctk.set_theme({'accent': '#e64980'})
```

Giving only the accent is enough: the hover color and the other keys that follow from it are derived. Unless you give 'accent\_hover' yourself, the hover color is derived from the accent: dark and medium accents are lightened (35% towards white), and very light accents are darkened (15% towards black), so the hover is always visible (see [Hover-variant](#)). The text on top of the accent color becomes black or white automatically, depending on how light the accent is. Existing widgets recolor immediately, and the accent stays when the appearance mode changes.

The dictionary `ctk.CANVAS_COLOR_THEMES` has five ready-made accents, 'blue' (the default), 'green', 'red', 'orange', and 'purple'. Each one is a theme dictionary, so you apply it with `set_theme`:

```
ctk.set_theme(ctk.CANVAS_COLOR_THEMES['green'])
```

### 10.3 Theme keys

The current theme is the dictionary `ctk.CANVAS_WIDGET_THEME`. It is built from the light or dark palette (`ctk.CANVAS_LIGHT_PALETTE`, `ctk.CANVAS_DARK_PALETTE`) plus the accent colors. These are its keys, and what the widgets use them for:

- `window`: the background of windows and of frames on windows.
- `surface`, `surface_hover`, `surface_disabled`: the background of entries, text boxes, list boxes, notebook pages, drop-downs, and unchecked boxes, in their normal, hovered, and disabled states.
- `neutral`, `neutral_hover`: unselected tabs and the segmented tab pill.
- `border`, `focus_ring`: normal and focused borders.
- `text`, `text_muted`, `text_disabled`: text, secondary text (arrows, accelerators) and disabled text.
- `track`, `thumb`, `thumb_hover`: slider and scroll bar tracks and thumbs.
- `separator`: separator lines, menu separators, and sash lines.
- `tooltip`: the tooltip background.
- `button_border`: the outline of buttons.
- `danger`: the red of the tab close button when hovered.
- `accent`, `accent_hover`, `accent_press`, `accent_text`: the accent color for buttons and highlights, its hovered and pressed versions, and the text on it.
- `selection`, `selection_text`: selected text in entries and text boxes.
- `row_current`: the highlight of the current row in lists and drop-downs.

You can change a key for *everything* by editing the dictionary before making your widgets (`ctk.CANVAS_WIDGET_THEME['surface'] = '##fafafa'`), but it is rebuilt when you call `set_appearance_mode` or `set_theme`; for lasting changes use `set_theme`, the named options, or a `theme` dictionary on the widgets.

## 10.4 Changing the theme dynamically

Every color is a theme key, and each level has methods to change keys while the program is running; everything affected recolors immediately, not only the accent. All of them take a dictionary of theme keys (see [Theme keys](#)). A value of `None` removes a change (and in a `theme` given when a widget or window is created, `None` simply means the default). If the dictionary has an `'accent'` (a `#rrggbb` color or a color name), the keys that follow from it (`accent_hover`, `accent_press`, `accent_text`, `focus_ring`, `selection`, `selection_text`, `row_current`) are derived automatically, unless you give them yourself.

From the widest to the narrowest level (a narrower level wins):

1. **The whole application:** `ctk.set_theme(theme)`, `ctk.reset_theme()`, `ctk.get_theme()`. The changes stay when the appearance mode is changed.
2. **One window:** `ctk.Window(theme = ...)` or `window.set_theme(theme)`, `window.reset_theme()`, `window.get_theme()` (also on `Toplevel`). They apply to the window and to every widget in it.
3. **One widget:** `widget.set_theme(theme)`, `widget.reset_theme()`, `widget.get_theme()` (and `theme` when creating it). This works on every canvastk widget, and on `Frame` and `ScrolledFrame` for their own background. A `Menu` and a `Tooltip` have `set_theme` too; it applies the next time they are shown.
4. **One color of one widget:** the named options such as `border_color` (see [Colors](#)).

**Example:**

```
import canvastk as ctk
root = ctk.Window()
b = ctk.Button(root, 'Hello')
b.grid()

ctk.set_theme({'surface': '#fffbe6', 'border': '#e67700'}) # every widget
root.set_theme({'window': '#fff3bf'}) # this window and its widgets
b.set_theme({'accent': '#e67700'}) # only this button
b.set_theme({'accent': None}) # remove that change again
print(b.get_theme()['accent'])
root.reset_theme()
ctk.reset_theme()
```

Since the palettes are dictionaries of theme keys too, you can make one window dark while the rest of the application is light:

```
root.set_theme(ctk.CANVAS_DARK_PALETTE)
```

**Note:** `set_theme` on a widget does not recolor the widgets inside it. Use the window's `set_theme` (or the global one) to recolor a whole window.

## 11 Functions

These are the functions and classes that the package exports besides the widgets:

- `set_appearance_mode`
- `get_appearance_mode`
- `set_theme`
- `get_theme`
- `reset_theme`
- `expand_theme`
- `hover_variant`
- `window_theme`
- `load_canvas_image`
- `make_font`
- `resolve_color`
- `parent_background`
- `rounded_box_image`
- `CanvasWidget`

### 11.1 `Set_appearance_mode`, `Get_appearance_mode`

See [Appearance and themes](#).

### 11.2 `Set_theme`, `Get_theme`, `Reset_theme`

`ctk.set_theme(theme)` changes theme keys for the whole application, `ctk.get_theme()` returns a copy of the current global theme, and `ctk.reset_theme()` removes the changes made with `set_theme`. See [Changing the theme dynamically](#).

### 11.3 `Expand_theme`

`ctk.expand_theme(theme)` returns the dictionary with the keys that follow from its 'accent' filled in. `set_theme` calls it for you.

## 11.4 Hover\_variant

`ctk.hover_variant(color)` returns a hover color for any tkinter color: lighter for dark and medium colors (35% towards white), darker for very light ones (15% towards black; a color counts as very light when its brightness is above 186 out of 255). It is what derives the hover color of an accent and of a button's `default_bg`.

## 11.5 Window\_theme

`ctk.window_theme(widget)` returns the theme dictionary of the window that contains the widget (empty if that window is not a canvastk Window or Toplevel). It is useful when writing your own widgets or popups.

## 11.6 Load\_canvas\_image

Use `ctk.load_canvas_image(path, size = None)` to load an image file as a `PhotoImage` that can be used for the `image` option of buttons, labels, and tabs. If `size` is `(width, height)` the image is resized to it with a high quality filter. Transparency is kept.

## 11.7 Make\_font

`ctk.make_font(font = None, size = 11, weight = 'normal')` returns a `tkinter.font.Font` from a `Font` (returned as it is), a font tuple or name, or `None` (the default family at the given size and weight). Fonts made this way are shared.

## 11.8 Resolve\_color

`ctk.resolve_color(widget, color)` converts any tkinter color (name, `#rgb`, `#rrggbb`) to `'#rrggbb'`.

## 11.9 Parent\_background

`ctk.parent_background(widget)` returns the background color of a widget as `'#rrggbb'`; it also works for ttk widgets.

## 11.10 Rounded\_box\_image

`ctk.rounded_box_image(width, height, radius, fill, outline = None, outline_width = 0, inset)` returns a cached, anti-aliased rounded box as a `PhotoImage` (transparent outside the shape). `fill` and `outline` are `'#rrggbb'` strings (`fill` may be `None`), `inset` shrinks the box on all sides, and `corners` says which of the top-left, top-right, bottom-right, and bottom-left corners are rounded.

## 12 Writing your own widget

`ctk.CanvasWidget` is the base class of every canvas-drawn widget. It handles the theme, the background, the colors, hover and focus tracking, and coalesces redraws (many changes in one event cycle cause one redraw). To make your own, subclass it and implement `redraw`:

```
class Badge(ctk.CanvasWidget):
    _COLOR_PARTS = ('badge_color',)

    def __init__(self, master, text = '', **kwargs):
        super().__init__(master, width = 80, height = 26, **kwargs)
        self.text = text
        self.track_interaction() # keeps self._hovered / self._focused

    def redraw(self, width, height):
        fill = self.part('badge_color',
            'accent_hover' if self._hovered else 'accent')
        self.draw_box(0, 0, width, height, fill, radius = height / 2)
        self.draw_content((0, 0, width, height), self.color('accent_text'),
            self.text)
```

Useful members:

- `self.part(name, default_theme_key)`: the widget's override named `name` if the user set it (through the `_COLOR_PARTS` names), otherwise the theme color `default_theme_key`. `self.color(key)` reads a theme key directly.
- `self.draw_box(x1, y1, x2, y2, fill, outline, outline_width, radius)` draws a rounded box. `self.draw_content(box, color, text, image)` draws an image and text side by side. `self.draw_chevron(x, y, direction, color)` draws an arrow head.
- `self.schedule_redraw()` requests a redraw. `self.refit()` recalculates the size from `requested_size()` (return `(width, height)` from it to make a widget that sizes to its content).
- `self.call_later(millisecons, function)` is after that is cancelled automatically when the widget is destroyed. `self.watch_variable(var)` redraws when a tkinter variable changes.
- `self.font, self.state (self.is_disabled()), self.fg, self.colors`.

## 13 Notes and limitations

- The popups of the [Menu](#), [Combobox](#), [OptionMenu](#) and [Tooltip](#) are borderless windows; their contents are drawn on a canvas, but their corners

are square, because a window cannot have transparent rounded corners everywhere.

- [Entry](#), [Textbox](#), [Spinbox](#) and [Combobox](#) use a native `tkinter.Entry` or `tkinter.Text` for the text itself. [Frame](#), [ScrolledFrame](#), [Panedwindow](#) and [Notebook](#) are tkinter frames (with canvas drawing for tabs, sashes and scroll bars), and `Window` and `Toplevel` are native windows. Everything else is drawn entirely on a canvas.
- A [Menu](#) is not a widget and cannot be attached to a window as its native menu bar; use a [MenuBar](#).
- Text on a canvas uses tkinter's normal text rendering; the shapes (boxes, tracks, thumbs) are anti-aliased, while thin lines like check marks and arrows are drawn with canvas lines.
- Creating hundreds of widgets takes a moment, because every widget is a Tk canvas. For long lists, use a [Listbox](#) (which draws only the visible rows) instead of one widget per item.